

Playful Frictional Design: A Design Space Grounded in Playfulness, Friction, and the Everyday

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Abstract

Friction in HCI has been explored as a means to provoke reflection and foster political engagement through design. Prior work has emphasized designs and participatory processes that introduce friction, encouraging reflection as a way to promote questioning and change of the status quo. We extend this scholarship by foregrounding conflict as an inevitable and necessary outcome of friction, asking how such conflict might be addressed constructively. Through theoretically engaging and discussing relevant literature, we propose playfulness as a design strategy to transform friction-induced conflict into constructive confrontation. We do so by articulating "Playful Frictional Design" as an approach grounded in theories of design friction and everyday playfulness. Finally, we outline directions for future design methods that leverage playfulness to create spaces for constructive conflict in design friction, aiming at promoting meaningful change.

CCS Concepts

• Human-centered computing → HCI theory, concepts and models.

Keywords

Playfulness, Friction, Agonism

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1 Introduction

In this paper, we argue for *playfulness* as an *approach for design friction*; an approach that not only accounts for conflict as a possible outcome of design friction, but proposes theoretical and practical means to engage with it *constructively*. In Human-Computer Interaction (HCI), friction has been used to promote reflection, encourage mindful behavior, and foster political engagement [8, 9, 15]. Although existing approaches have proposed theoretical and practical implications of designing for friction, the role of conflict in the process of reflection and contestation has been less explored. We propose *Playful Frictional Design* as an approach that embraces conflict as a necessary step to transform reflection into meaningful change in frictional design.

Building on Sicart's proposal of agonism as a framework to understand political play [22], we considered how conflict could be reframed as a core starting point that stimulates reflection and challenges assumptions. Agonism is a theoretical standpoint that frames conflict as an essential and productive component of democratic political engagement [17]. Sicart argues that playfulness creates a way of engaging with everyday life that opens opportunities for political engagement, which he frames through the lenses of agonism. In connecting these ideas with the notion of friction, we articulate how design can foster inclusion, plurality, and empowerment as outcomes of constructive conflict (that is, agonistic conflict, under the lenses of agonism [17]). Playfulness comes in as a practical approach to design spaces (of reflection, discussion, and interaction) where the conflict resulting from friction can develop constructively.

In our contribution, we introduce the theoretical foundations and the design space of Playful Frictional Design. This design space is defined by: (1) *the frictional*, based on the definition of frictional design by Pierce [18]; (2) *the playful*, a definition and approach for everyday playfulness based on Situated and Emergent Play by Altarriba Bertran et al. [2]; and (3) *the everyday*, the understanding of everyday in design friction [15].

Through reflecting on related work on design friction and playfulness in HCI, we make visible both their theoretical overlaps and important distinctions. This paper should be understood less as a



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fully realized and evaluated design framework, and more as an argument piece that delineates the design space of Playful Frictional Design. Our contribution lies in providing a theoretical grounding and definition that connects design friction, playfulness, and the concept of everyday. Finally, we discuss how researchers can apply, validate, and evaluate our work through case studies in future research. With this contribution, we aim to establish the foundations for future inquiry into how playfulness can serve as an approach to create spaces for agonistic conflict in design friction.

2 Theoretical Foundations

This section explores the theoretical foundations of Playful Frictional Design by examining core HCI literature on design friction and everyday playfulness. It highlights where these concepts intersect, where they differ, and identifies gaps for further exploration in Playful Frictional Design.

2.1 Design Friction in HCI

Friction, or design friction, is widely established in HCI as a means to promote reflection or shift user behavior. It has been used to break seamless interface design and prompt reflection, for example: in autonomous vehicles to increase awareness during takeover from autonomous to manual drive [3], in digital payment to foster conscious budgeting [14], and to counteract doomscrolling [19].

Forlano and Mathew [12] expanded friction from small interface disruptions to participatory design, defining it as *a way to highlight controversies and push socio-technical discussions*. In Adversarial Design, DiSalvo [8] built on agonism to frame friction as a *design strategy to provoke contestation, challenge norms, and invite political debate through artifacts*. Hassenzahl and Laschke [13] proposed an aesthetics of friction based on artifacts designed to foster behavior change in everyday life.

Building on this, Korn and Volda [15] situated friction in civic engagement to provoke people to question norms by taking active roles in their communities. Their contribution is relevant here, as they ground the situatedness of friction in theories of everyday life, which not only conceptually connects to everyday playfulness, but also helped us understand the relevance of contextualization in Playful Frictional design (discussed in Section 3.3).

These examples fall under what Pierce [18] called frictional designs. He argued that traditional definitions of design align with progressional design, converging toward production (e.g., design thinking), whereas frictional designs have transproductional value, producing outcomes such as reflection, social debate, or change. In section 3.1, we further elaborate on how *the frictional* shapes our notion of Playful Frictional Design.

2.2 Everyday Playfulness

Contemporary game and play literature increasingly understands play as embedded in everyday life, shaped by and shaping social norms, identities, and practices rather than existing apart from them. From this perspective, play can be more than purely recreational and participate in broader cultural and political discourse. Bogost [4] articulates this through persuasive games and the notion of *procedural rhetoric*, where rules, systems, and mechanics communicate ideas and influence players' understanding and actions.

Flanagan and Nissenbaum [11] further position play as a *tool for reflection and critique*, arguing that gameplay can be designed to interrogate values, power structures, and societal norms encountered in everyday life. Through critical and radical game design, designers can embed these concerns directly into rules, narratives, and mechanics, ensuring that play itself invites reflection and challenges assumptions [10]. Seen this way, play is inherently connected to the everyday and holds potential as a medium for critical and political engagement beyond entertainment.

Inspired by play's potential to influence everyday life, Sicart proposed *agonism as a framework to understand the political potential of play* [22]. For him, playfulness, the attitude of play, carries this political potential. By framing play as contestation (involving negotiation and conflict with rules, opponents, or systems) and as political engagement (foregrounding disagreement and power structures rather than smoothing them over), he presents an agonistic perspective on everyday play that can challenge routines, question systems, and enable critical reflection.

Similarly exploring play's connection to the everyday and its potential to foster well-being rather than pure entertainment, Altarriba Bertran et al. framed play as a *lived, situated, and emergent phenomenon*, deeply intertwined with everyday practices and environments [1, 2]. They argue that playfulness cannot be fully designed or prescribed, but offer guidance on how designers can scaffold conditions under which it emerges, a design process they call Chasing Play [1]. In Section 3.2, we elaborate on how everyday playfulness theoretically grounds Playful Frictional Design.

3 Design Space of Playful Frictional Design

In Playful Frictional Design, we aim to provide a theoretical foundation and definition of concepts for a design space that explores how playfulness can create opportunities in which conflict arising from friction develops constructively, ultimately prompting reflection and questioning of the status quo. Figure 1 illustrates how we envision the relationship between the grounding literature for these concepts. We summarize the three foundational concepts for Playful Frictional Design below and expand their definition in the next sections.

The frictional: a frictional design that embraces agonistic conflict as a design goal.

The playful: playfulness as the attitude of play that manifests in everyday life, used to create social and emotional well-being, rather than productivity or pure enjoyment.

The everyday: theoretical and practical implications of situating friction and playfulness for political engagement in the product-residue of everyday life.

The design space we propose does not develop under existing streams of frictional design. Instead, we establish it *in conversation with these approaches* as a different way of creating friction through design. Like Adversarial Design [8] and Critical Design [9], our approach uses artifacts to foster reflection and invite political debate. However, we go further to *embrace agonistic conflict as a result of friction* and as a necessary step for transforming reflection into change. Our process makes designing for agonistic conflict a design goal, ensuring that the resulting conflict is constructive.

In contrast to the Aesthetics of Friction [13], our approach does not aim at persuasion or behavior change. We align more closely with participatory approaches [12, 15] in making space for democratic engagement in everyday life. The difference lies in design practice: we start from a play design method and center agonistic conflict as a design goal to create spaces for agonism.

3.1 The Frictional

Pierce [18] defined frictional designs (e.g., Speculative Design, Critical Design) as "alternative" designs that, in contrast to "conventional" designs (e.g., Human-Centered Design), do not aim to converge toward production. Both types of design make use of actualizations of possible futures to produce value. In the case of frictional designs, such actualizations don't need to necessarily be complete in a way that they can be produced. They can be used for reflection, speculation, critique, discourse, agonism, etc. That is why he positions these "alternative" designs as ones that generate friction, hence calling them frictional designs.

Within Playful Frictional Design, we embrace agonistic conflict as a productive and necessary outcome of frictional design. Agonistic conflict comes from the political framework of agonism, which views conflict as essential to fair and plural democratic processes [17]. Rather than eliminating opposing views, agonism places *constructive conflict as a step toward democratic change*, valuing opposing voices and diverse perspectives. In setting agonistic conflict as a design goal, we enrich our conceptual foundation and design approach by adding strategies to create spaces for agonistic conflict and avoid antagonism as an outcome. Consequently, friction can take a step from reflection toward actionable change by sustaining contestation while recognizing the legitimacy of differing perspectives.

3.2 The Playful

Designing for friction can create discomfort by pushing boundaries, fostering reflection, and prompting confrontation, outcomes that do not always align with common notions of fun, enjoyment, or amusement. Sicart proposed playfulness as the *attitude of play*, in contrast to play, the activity [21]. We adopt Sicart's definition, as it fits a view of the attitude of play that goes beyond entertainment.

Aligning with the non production-oriented value of friction, our motivation for using playfulness within Playful Frictional Design is eudaimonic: aiming to be socially and emotionally productive rather than materially productive or purely enjoyable [20]. Other motivations of play (problem-oriented or hedonic [6, 20]) can create friction but must be carefully designed to reflect the principles of Playful Frictional Design, whereas eudaimonic play offers a more fitting conceptual definition.

We use Situated and Emergent Play [2] to frame our understanding of everyday playfulness. By observing everyday contexts and identifying play potentials, it inspires playful design that emphasizes emotional value and well-being rather than performance or productivity, aligning with eudaimonic play and the transproductive value of friction.

3.3 The Everyday

Following Korn and Volda [15], we base our interpretation of everyday life on Lefebvre [16] and de Certeau [7]. Lefebvre sees everyday life as the space where political engagement must be present for transformation. The everyday is both a product and a residue of routine, labor, consumption, or leisure, shaped by tensions between personal and social. In this scenario, political engagement is often reserved for "privileged moments," so true citizenship requires acting within the *product-residue* beyond voting or protests, re-appropriating time, space, relationships, and objects to infuse it with new values, rhythms, and practices. De Certeau shifts focus from structures of power to subtle actions of resistance and subversion in everyday life, distinguishing *tactics*, ways people resist (e.g., pirating music), from *strategies*, institutional means to enforce dominance (e.g., cities, armies). Everyday practices occur in the cracks of official systems, reinforcing Lefebvre's *product-residue* as a space for resistance through *tactics*.

We understand that a practice rooted in everyday life *must be tried, lived, and experienced in context*. In our case, design friction and playfulness are deeply embedded in people's everyday life. This means that in addition to creating spaces for tactics to emerge in the *product-residue*, this approach also requires empirical work.

4 Discussion and Future Work

We propose Playful Frictional Design as a new design space that opens possibilities for using playfulness as an approach to create spaces for agonistic conflict. In this section, we share our vision for future work to develop this design space: (1) **bridging theory and practice** by developing a design process, (2) **validating and evaluating** the theoretical framing through case studies, and (3) a **reflection on the challenges** we faced in starting to define suitable real-life contexts for the case studies.

4.1 A Design Process for Playful Frictional Design

Altarriba Bertran et al. introduced Chasing Play [1] as a method for Situated and Emergent Play, expanding how HCI understands and designs for play, especially beyond formal games. Campreguer França et al. [5] built on Chasing Play to propose a process that uses everyday playfulness to create spaces for agonism. They do so by adding steps into Chasing Play that center the principles of agonism as a design goal, principles originally identified by Willis [23] in his work on bridging agonism and design practice. The process in [5] starts by (1) **tracing frictions** to identify latent frictions that ground the design, (2) **chase the play** by observing and engaging with play potentials, (3) **enhance the play** by designing playful interactions that leverage these potentials and integrate agonistic principles, and (4) **deploy the play** by situating the intervention in the original setting. Frictions and play potentials together provide a *contextual foundation for design*, while the principles of agonism are operationalized as *actionable design elements and guiding questions* for data collection and analysis [6]. Although not exactly framed as such, the work from Campreguer França et al. [5] well illustrates what we envision as a starting point to develop a design process for Playful Frictional Design. In the next paragraphs, we discuss how

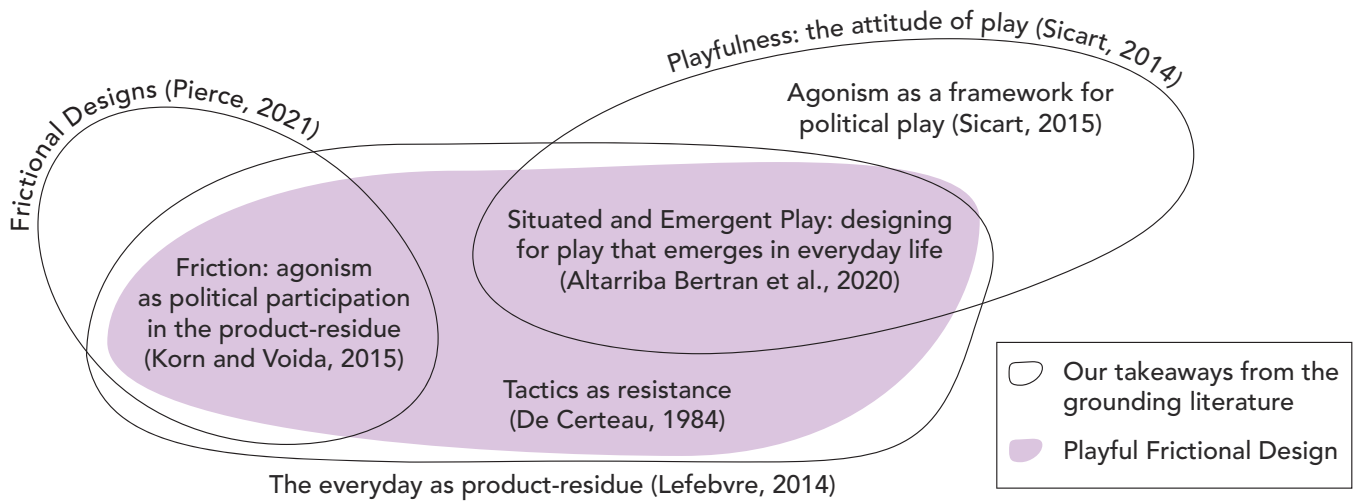


Figure 1: Overview of the literature grounding the design space of Playful Frictional Design. Note that the relations established here show how we connect the concepts and do not always reflect the original references of the authors. For example, Altarriba Bertran et al. [2] does not frame everyday under Lefebvre [16], but we understand it as fitting under this definition.

this process can be further developed to better fit our proposed design space.

In contrast to prior frictional design approaches, where designers define the friction based on their own interpretations, we argue for actively engaging participants in "tracing frictions" through participatory or co-design methods. Rather than starting from a preset topic, as in Campreguer França et al. [5], this approach can foster plurality in contestation by opening up spaces of conflict that participants can appropriate. A playful perspective further supports openness, curiosity, and engagement throughout the design process. Regarding evaluation, we argue that the process can benefit from an evaluation method that targets the principles of agonism and provides a comparative base for other case studies, something that was not clearly discussed in Campreguer França et al.'s implementation. For example, a great starting point could be Willis' discussion on his use of Discourse Analysis and game data (data collected from participants in a game setting) to collect proof that the strategies for agonism succeeded. He points out that even though (Critical) Discourse Analysis helps investigate hegemonies and imbalances of power through the lenses of agonism, the data extracted "must be acknowledged as constructions, influenced and selected by a creative practitioner" [23, p 125]. This reflection is something we aim to explore in future developments of Playful Frictional Design. Regarding game data, Willis explains that based on his experiments, it does reflect participants real perspectives and values, even though the data is collected in a sort of in-between the real world context and the game imaginary, or a context created through the game itself—agreeing with the understanding of procedural rhetoric [4].

4.2 Validate and Evaluate in Practice Through Case Studies

We invite future work to validate and evaluate the theoretical grounding and definition of the design space we propose in this paper by implementing the design process from Campreguer França et al. [5], while situated in our proposed foundation for Playful Frictional Design. This means validating the suitability of the theoretical framing and the defining concepts of the design space when applying them into practice. For example, do the definitions of *the frictional*, *the playful*, and *the everyday* fully describe what is observed when implementing Playful Frictional Design? What are the limitations of Playful Frictional Design? Does it fit better in certain contexts than in others? Is it best suited for a specific target audience?

4.3 Reflections and Challenges in Approaching Case Studies

We are working to define suitable research contexts for validating and evaluating the design space proposed in this paper, a process that already raised certain limitations and questions. Currently, we are approaching relevant institutions and have encountered barriers from their side, such as a lack of time and resources to engage with the project, as well as difficulties in seeing the benefit due to the exploratory nature of the methods. We find it important to share our experience because it connects back to Willis' notes on how the "critical" in his use of Critical Discourse Analysis was defined by the researchers [23, p 125]. In our context, the reflection concerns who initiates the process of creating spaces for agonism. Looking ahead, a key question is how to foster practices that motivate individuals to develop their own tactics, rather than relying on interventionist methods. Given challenges in securing external partners, we proceeded by planning a study focused on a friction identified within the first author's university (AI in teaching). Simultaneously, we

have explored applying this approach to post-colonial perspectives in museum exhibits. This latter initiative has only been feasible through collaboration with interested colleagues from the history department (again, the university context), who view the partnership as an opportunity to secure funding and connect their research with practice.

In sum, the theoretical framing and methodological discussion presented in this work should spark discussion and discourse on the intersection of playful approaches for political engagement in everyday life and trigger the next steps in validating and evaluating the proposed design space.

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